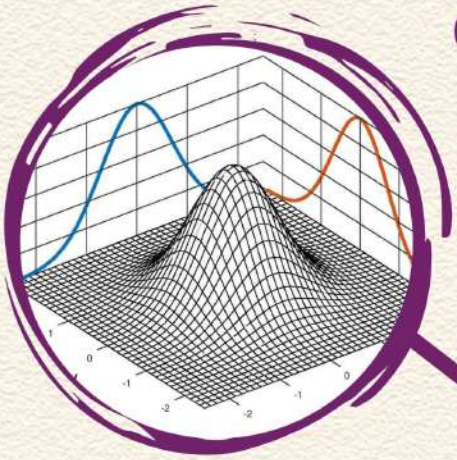
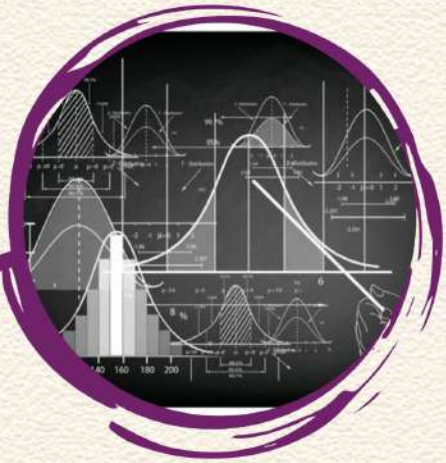
Spark
POLYTECHNIC

إحصاء واحتمالات

ملخص قوانين







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0782032213 طلاب البور والإتصالات _ البوليتكنك

- $p(A) = \frac{\#\text{ of elements in } A}{\#\text{ of elements in } \Omega}$ ↔ impossible event: $p(A)=0$; ordinary event: $0<p(A)<1$; sure event: $p(A)=1$

- # of Ω in the experiment of
  - > tossing a coin n times 2^n.
  - > rolling a dice n times 6^n.
  - > in general $nCr = \binom{n}{r} = n!/r!(n-r)!$.
- $p(A \cap B)$ → A, B independant: $p(A).p(B)$
  - → A, B disjoint: 0

  intersection keywords: and / both.
- $p(A \cup B)$ → A, B independant: $p(A)+p(B)-p(A \cap B)$
  - → A, B disjoint: $p(A)+p(B)$

  union keywords: or / at least.
- $p(\bar{A}) = 1 - p(A)$, $\bar{\bar{A}} = A$
  - > $\bar{A} \cap A = \Omega \longrightarrow p = 1$
  - > $\bar{A} \cup A = \phi \longrightarrow p = 0$
- demorgans Law $\cup \leftrightarrows \cap$ وزع المتممة وغير الإشارة

$\bar{A} \cap \bar{B} = \overline{A \cup B} \longrightarrow p(\bar{A} \cap \bar{B}) = p(\overline{A \cup B}) = 1 - p(A \cup B)$

$\bar{A} \cup \bar{B} = \overline{A \cap B} \longrightarrow p(\bar{A} \cup \bar{B}) = p(\overline{A \cap B}) = 1 - p(A \cap B)$

- exception Law

$p(A-B) = p(A \cap \bar{B}) = p(A) - p(A \cap B)$

لو صيغة ضرورة | لو صيغة غير ضرورة | احتمال الي طالعلي متممة | احتمال الي بحت المتممة

- Conditional probability (knowing/such/given that)

$p(A|B) = \frac{p(A \cap B)}{p(B)}$ = احتمال التقاطع / احتمال الشرط

if A, B independant $p(A|B) = \overline{p(A)}$

> $p(\bar{A}|B) = 1 - p(A|B)$

> $p(\phi|A) = 0$

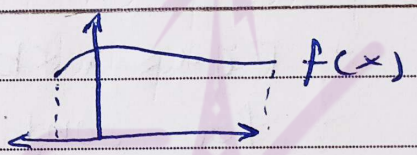
> $p(\Omega|A) = 1$

> $p(A|\Omega) = p(A)$

• Random Variables :-

	discrete	continuous
values	$x = \{\ ,\ \}$	$x \in [\ ,\ ]$
	set / subset	interval

• probability density function $\left(\frac{dF(x)}{dx}\right)$:-

	discrete	continuous
probability	$p(x)$	$f(x)$
	$0 \leq p(x) \leq 1$	$f(x) \geq 0$
p.d.f shape	$\boxed{x \mid p(x)}$ table with values	 cont. function above x-axis
Area	$\sum_{x \in R_x} p(x) = 1$	$\int_{R_x} f(x)\,dx = 1$
Expectation EX	$\sum_{x \in R_x} x\,p(x)$	$\int_{R_x} x\,f(x)\,dx$

> Variance : $EX^2 - (EX)^2$, $Var(x) \geq 0$
for both $Var(ax+b)$, $a^2\,Var(x)$

> std : $\sqrt{Var(x)}$, $std(x) \geq 0$
for both $std(ax+b)$, $|a|\,std(x)$

> expectation : $E(ax^2+bx+c)$
for both $= aEX^2 + bEX + c$

* ملاحظة : أو يساوي ما بتفرق بال continuous أما discrete بتفرق!

• complutative distribution function (F(x)) :-

	discrete	continuous
c.d.f	$\sum_{n<x} p(u)$	$\int_{-\infty}^{x} f(u)\,du$
shape of c.d.f	step function	continuous function

properties of c.d.f :

* $F(-\infty) = 0$, $F(\infty) = 1$

* $F(a \leq x \leq b) = F(b) - F(a)$
for continuous random variable.

* $F(x_1) \leq F(x_2) \iff x_1 \leq x_2$

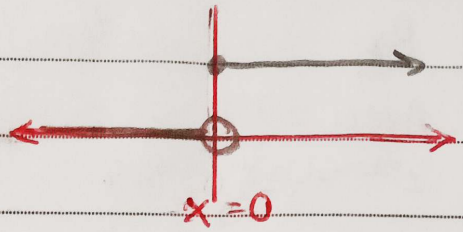
* F is increasing in both cases.

		discrete	continuous
joint p.d.f	R_{xy}	ordered pairs	region
	Area المساحة	$\sum_{x\in R_x}\sum_{y\in R_y} f(x,y) = 1$	$\iint_{R_{xy}} f(x,y)\,dx\,dy = 1$
	EXY (+/0)	$\sum_{x\in R_x}\sum_{y\in R_y} xy\,f(x,y)$	$\iint_{R_{xy}} xy\,f(x,y)\,dx\,dy$
	$Eg(x,y)$	$\sum_{x\in R_x}\sum_{y\in R_y} g(x,y)\,f(x,y)$	$\iint_{R_{xy}} g(x,y)\,f(x,y)\,dx\,dy$
	$Cov(x,y)$	$EXY - (EX)(EY)$ (+/0/-)	
	$Cov(ax+b, cy+d)$	$ac\,Cov(x,y)$ (+/0/-)	
	$Corr(x,y)$	$\frac{Cov(x,y)}{\sigma_x \sigma_y}$ $[-1,1]$	
	$Corr(ax+b, cy+d)$	same sign for a & c	$+ Corr(x,y)$
		diff. sign for a & c	$- Corr(x,y)$
marginal p.d.f of X	$f_x(x)$	$\sum_{y\in R_y} f(x,y)$	$\int_{R_y} f(x,y)\,dy$
	EX	$\sum_{x\in R_x} x\,f_x(x)$	$\int_{R_x} x\,f_x(x)\,dx$
marginal p.d.f of y	$f_y(y)$	$\sum_{x\in R_x} f(x,y)$	$\int_{R_x} f(x,y)\,dx$
	EY	$\sum_{y\in R_y} y\,f_y(y)$	$\int_{R_y} y\,f_y(y)\,dy$

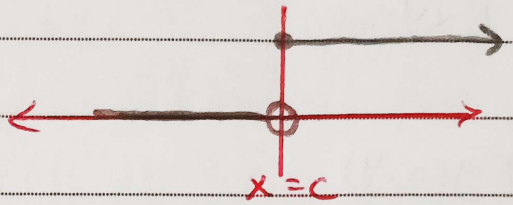
- Remmember: if X, Y are independant r.vs then:
  - $f(x,y) = f_x(x) \cdot f_y(y)$
  - $E(x,y) = EX \cdot EY$
  - $Cov(x,y) = 0$ & $Corr(x,y) = 0$

- Unit step function:

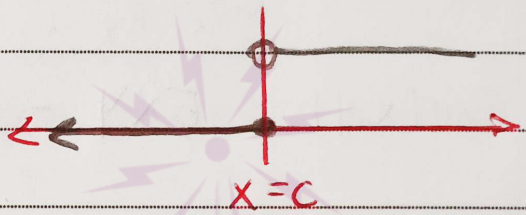
$$u(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$



$$u(x-c) = \begin{cases} 1, & x \geq c \\ 0, & x < c \end{cases}$$



$$u(c-x) = \begin{cases} 0, & x > c \\ 1, & x \leq c \end{cases}$$



- Impulse function:

$$- \int_{-\infty}^{\infty} \varphi(x)\, \delta(x-c)\, dx = \varphi(c)$$

$$- \int_{-\infty}^{\infty} \delta(x-c)\, dx = 1$$

- Properities:

$$- \frac{d\,u(x)}{dx} = \delta(x) \qquad \frac{d\,F(x)}{dx} = f(x)$$

- for discrete r.v X

c.d.f x:

$$F(x) = P_r(x_1)\, u(x-x_1) + \dots + P_r(x_n)\, u(x-x_n)$$

p.d.f x:

$$f(x) = P_r(x_1)\, \delta(x-x_1) + \dots + P_r(x_n)\, \delta(x-x_n)$$

Discrete Distributions

	Binomial	Poisson	Additional law in poisson dist.:-
الرمز	$X \sim B(n,p)$	$X \sim P(m)$	$EX^2 = m^2 + m$
p.d.f $p(x)$	$\binom{n}{x} p^x q^{n-x}$	$\frac{e^{-m} m^x}{x!}$	
values of x	$0 \longrightarrow n$	$0 \longrightarrow \infty$	⇝ positive integers
Var(X)	$Var(X) = qEX = qpn$	$Var(x) = EX = m$	
parameters	n, p, q (3)	m (1)	

Continuous Distribution

	Normal	Std Normal	Uniform
الرمز	$X \sim N(\mu, \sigma^2)$	$z \sim N(0,1)$	$X \sim U(a,b)$
p.d.f	-	-	$f(x) = \begin{cases} \frac{1}{b-a}, & a \le x \le b \\ 0, & o.w \end{cases}$
c.d.f	$F(x)$	$T(x)$	-
values of r.v	$-\infty \longrightarrow \infty$	$-\infty \longrightarrow \infty$	$a \longrightarrow b$ ⇝ Real numbers
parameters	μ, σ (2)	μ, σ (2)	a, b (2)

$z = \frac{x-\mu}{\sigma}$

Expected Values of functions of r.vs:-

- x, y are indep. : $R_{xy} = (EX)(EY)$.
- x, y are orthogonal : $R_{xy} = zero$.

$x(t, s)$

Random process → Sample function $s = s_0$

Random process → Random Variable $t = t_0$

wide-sense stationary:
- 1st order: $E(x(t)) = \bar{X}$
- 2nd order: $R_{xx}(\tau)$

• Correlation Function

> Auto Corr: $R_{xx}(t, t+\tau) = E(x(t) \cdot x(t+\tau))$

$R_{xx}(\tau = 0) = E(X^2)$

$\lim_{|\tau| \to \infty} R_{xx}(\tau) = (EX)^2 \longrightarrow EX = \sqrt{\lim_{|\tau| \to \infty} R_{xx}(\tau)}$

* $Var(X) = R_{xx}(\tau = 0) - \lim_{|\tau| \to \infty} R_{xx}(\tau)$

> Cross Corr: $R_{xy}(t, t+\tau) = E(x(t) \cdot y(t+\tau))$

jointly wide-sense: $R_{xy}(\tau)$

orthognal r.p: $R_{xy}(\tau) = 0$

indepdant r.p: $E(x(t)) \cdot E(y(t+\tau))$

indep. + wide r.p: $\bar{X}\bar{Y}$

• Covariance Function

> Auto Cov: $C_{xx}(t, t+\tau) = R_{xx}(t, t+\tau) - E(x(t)) \cdot E(y(t+\tau))$

$C_{xx}(t, t+\tau)\big|_{\tau=0} = Var(X)$

> Cross Cov: $C_{xy}(t, t+\tau) = R_{xy}(t, t+\tau) - E(x(t)) \cdot E(y(t+\tau))$

$x(t), y(t)$: $C_{xy}(t, t+\tau) = 0$

both indep.

* المتطابقات المثلثية :

- $\cos(a+b) = \cos a \cos b - \sin a \sin b$
- $\cos(a-b) = \cos a \cos b + \sin a \sin b$

متطابقات مطلوب منك حفظها

- $\sin(a+b) = \sin a \cos b + \cos a \sin b$
- $\sin(a-b) = \sin a \cos b - \cos a \sin b$

- $\cos(-x) = \cos(x)$
- $\sin(-x) = -\sin(x)$

متطابقات تعطى لك بالامتحان (مش محفوظ)

- $\dfrac{\cos(a-b) + \cos(a+b)}{2} = \cos a \cos b$
- $\dfrac{\cos(a-b) - \cos(a+b)}{2} = \sin a \sin b$